



Differentiable design of progressive-addition lens using NURBS surface

XINGHUA PAN,^{1,2} HAISONG TANG,^{1,2} ZEXIN FENG,^{1,2,*} AND HUAZHONG XIANG³

¹Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, Beijing 100081, China

²MOE Key Laboratory of Optoelectronic Imaging Technology and Systems, Beijing Institute of Technology, Beijing 100081, China

³School of Medical Instrument and Food Engineering, University of Shanghai for Science and Technology, Shanghai 200093, China

*fzx84@126.com

Abstract: Presbyopia is an age-related condition that gradually reduces the eye's ability to focus on nearby objects, affecting over a billion people globally. The progressive-addition lens (PAL), commonly used for presbyopia correction, struggles to eliminate inherent astigmatism in the lateral areas of the progressive corridor. In this paper, we propose an efficient method for designing PAL using a differentiable optimization approach based on non-uniform rational B-spline (NURBS) surface, with an initial estimate derived through a modified variational-difference numerical method. Our method focuses on addressing a nonlinear error function without local approximations. In the two design examples, the high-level astigmatism areas near the intermediate transition zone are pushed toward the edge of the lens, demonstrating the effectiveness of our method.

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1. Introduction

Presbyopia is essentially a decline in the physiological accommodative power of the eye, typically occurring after the age of 40 [1]. The prevalence of presbyopia and its impact on visual impairment are increasing, especially in countries like China [2], due to the aging population. Presbyopia can be corrected by wearing framed spectacles, using contact lenses or undergoing surgery [3]. Among these options, wearing spectacles is the most common approach owing to its convenience and safety. Notably, progressive-addition lens (PAL) [4] has attracted public attention for providing a continuous variation in optical power across the effective visual region.

As illustrated in Fig. 1, PAL can be divided into four areas: the far zone with low power for distance vision, the near zone with high power for close-range activities, the intermediate transition zone with progressive power connecting the far zone and the near zone, and the blending zone which is distributed along both sides of the lens. The design of PAL mainly encompasses direct method and indirect method [5]. In the direct method studied by Winthrop [6], a prescribed polynomial distribution is used to represent the curvature variation along the vertical meridian. Horizontal curves across the meridian are chosen to have the same curvatures at their intersection points with the meridian. Savio et al. [7] proposed a parametric method for geometric modelling of PAL, which enables the control of the addition power, the corridor length, etc. By discretizing curves with known curvature, PAL can be designed to suit the individual characteristics of the wearer. Compared to using the analytical solution of the Laplace equation to obtain the level curves, Tang et al. [8] employed a numerical method to solve the Laplace equation, with boundary and link conditions determined by personal requirements. These aforementioned direct design methods focus on the assignment of desired power, with little control of astigmatism over the lens.

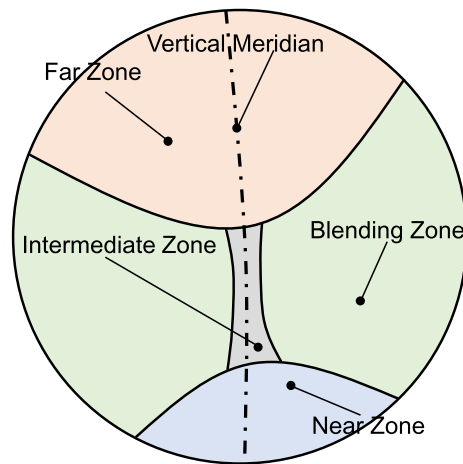


Fig. 1. Schematic of PAL zones.

The principle of the indirect design method is to minimize astigmatism while achieving the best possible match to the desired power distribution. Loos et al. [9] minimized an error functional comprising the deviation from the prescribed optical power and the unwanted astigmatism, each with different respective weights. Thus, the design of PAL can be posed as a variational problem, which can be transformed into a nonlinear fourth-order partial differential equation (PDE) known as the Euler-Lagrange equation [5]. Wang et al. [5,10] utilized the finite element method to solve a linear fourth-order PDE by assuming that the desired surface is a slight perturbation of a spherical surface. Jiang et al. [11] proposed a variational-difference method to design the PAL. They solved a linearized functional minimization problem using the trapezoid quadrature rule and the finite difference method. Li et al. [12] employed a customized frame function to design the PAL with variational-difference method, achieving low levels of astigmatism for wearers. However, most of the indirect design methods rely on the approximation to the error functional that is quadratic. The final solution of the variational problem could be sub-optimal. Additionally, the numerical methods they employed for designing PAL are inefficient, primarily due to the generation of meshes (especially on irregular boundaries) and the computation of derivatives.

Instead of solving a quadratic error functional or a linear fourth-order PDE, we employ non-uniform rational B-spline (NURBS) surface to optimize a highly nonlinear error functional without local approximations. Our method effectively optimizes both the desired power distribution and the unwanted astigmatism within the effective visual area. The high-level astigmatism areas near the intermediate transition zone are pushed towards the edge of the lens, making it more acceptable and comfortable to wearers. Section 2 describes the detailed differentiable method using NURBS surface. Section 3 gives two design examples to demonstrate the effectiveness of the proposed method. In Section 4, we give a short conclusion to this paper.

2. Method

The proposed design method of PAL is based on the minimization of a nonlinear error functional. Figure 2 illustrates the procedures of our method. To improve the optimization efficiency, we obtain an initial estimate of PAL using a modified variational-difference numerical method [11], which solves a linear fourth-order PDE with respect to the perturbation term directly instead of approximating the solution of a quadratic functional. After utilizing NURBS surface to fit the discrete points from initial estimate, the nonlinear error functional can be built with automatic differentiation (AD) technique. Finally, we optimize the NURBS surface to minimize the error

functional by selecting an appropriate weight ratio for optical power and astigmatism. The optical properties including sphere power and cylinder power are evaluated with contour maps.

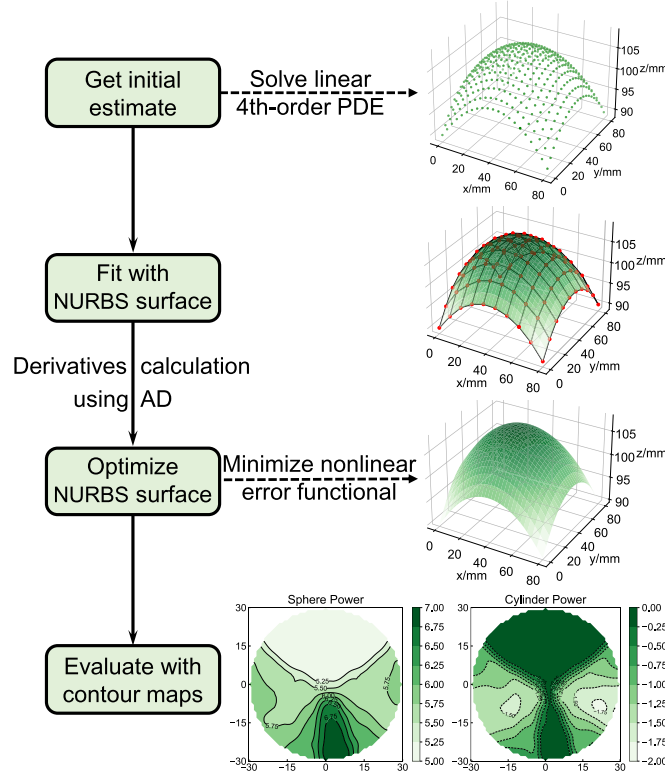


Fig. 2. Flow chart of our design method for PAL.

2.1. Functional minimization problem

Optical power is a crucial design parameter to PAL, which is determined by the mean curvatures of the front surface (P^f) and back surface (P^b). In ophthalmic optics, the optical power P of lens is given by the formula [5,9]

$$P = (1 - n)P^b + \frac{(n - 1)P^f}{1 - d \left(1 - \frac{1}{n}\right)P^f}, \quad (1)$$

where n is the refractive index of lens, d is the central thickness of lens. Suppose k_1 and k_2 are the two local principle curvatures at any surface, the values of mean curvature μ and curvature difference δ can be obtained by

$$\begin{cases} \mu = \frac{k_1 + k_2}{2} \\ \delta = |k_1 - k_2| \end{cases}. \quad (2)$$

Then the mean power P_m and the astigmatism P_{ast} (also known as cylinder power) are given by

$$\begin{cases} P_m = 1000(n - 1)\mu \\ P_{ast} = 1000(n - 1)\delta \end{cases}, \quad (3)$$

measured in diopter (D).

The core function of PAL is providing appropriate optical power in patient's distance and reading visions, with progressive and smooth shift in intermediate transition zone. Another key consideration in PAL design is the control of astigmatism, as it negatively affects the visual quality experienced by wearers. Because of the relationship between the mean power and the astigmatism, as shown in Eq. (2), the optical properties of PAL must be balanced on demand during the practical design process. Loos et al. [9] constructed an error functional to satisfy both of the design requirements, as formulated by

$$I(z) = \int_{\Omega} \left\{ \alpha(x, y) P_{\text{ast}}(x, y)^2 + \beta(x, y) (P_m(x, y) - P_0(x, y))^2 \right\} dx dy. \quad (4)$$

Here, z is the sag distribution of PAL, $P_{\text{ast}}(x, y)$ and $P_m(x, y)$ are the designed values of astigmatism and mean power, $\alpha(x, y)$ and $\beta(x, y)$ are the two positive weight distributions (or matrixes) of $P_{\text{ast}}(x, y)$ and $P_m(x, y)$ respectively, $P_0(x, y)$ is the desired optical power, and Ω is the computational domain. The determination of the values of $\alpha(x, y)$ and $\beta(x, y)$ depends on the values of astigmatism and optical power at the point (x, y) in PAL. This means that the weight $\alpha(x, y)$ will be greater in areas where the lens needs to minimize astigmatism, and similarly, the weight $\beta(x, y)$ will be larger in regions where the lens must meet the prescribed power. After plugging Eq. (2) and Eq. (3) into Eq. (4) and removing constant terms, we obtain

$$I(z) = \int_{\Omega} \left\{ \alpha(x, y) \left(\frac{k_1(x, y) - k_2(x, y)}{2} \right)^2 + \beta(x, y) \left(\frac{k_1(x, y) + k_2(x, y)}{2} - \mu_0(x, y) \right)^2 \right\} dx dy. \quad (5)$$

Here, $\mu_0(x, y)$ represents the desired mean curvature distribution. For a given sag distribution $z(x, y)$, the mean curvature μ and the Gauss curvature $\kappa = k_1 \cdot k_2$ can be expressed as

$$\begin{aligned} \mu(x, y) &= \frac{(1+z_x^2)z_{yy} - 2z_x z_y z_{xy} + (1+z_y^2)z_{xx}}{2(1+z_x^2 + z_y^2)^{3/2}} \\ \kappa(x, y) &= \frac{z_{xx}z_{yy} - z_{xy}^2}{(1+z_x^2 + z_y^2)^2}. \end{aligned} \quad (6)$$

Next, substitute k_1 and k_2 with μ and κ using Eq. (2) and Eq. (6). Then Eq. (5) can be converted into

$$I(z) = \int_{\Omega} \left\{ \alpha(x, y) (\mu(x, y)^2 - \kappa(x, y)) + \beta(x, y) (\mu(x, y) - \mu_0(x, y))^2 \right\} dx dy. \quad (7)$$

The goal of PAL design is to determine the optimal sag distribution $z(x, y)$ that minimizes the error functional $I(z)$.

2.2. Initial estimate of PAL

It is difficult to minimize the error functional $I(z)$ directly, so we get an initial estimate of PAL by solving a linear fourth-order PDE which only takes a few seconds. The minimization of the error functional $I(z)$ can be converted into solving a nonlinear fourth-order PDE. Wang et al. [10] linearized the nonlinear PDE by assuming that the designed PAL $z(x, y)$ is a combination of a given spherical background surface $w(x, y)$ and a slight perturbation $o(x, y)$. Then Jiang et al. [11] applied a variational-difference method to solve a quadratic functional based on the linearization approximation method. Following the linearization approach proposed by Wang et al. [10], substituting $z(x, y) = w(x, y) + o(x, y)$ into Eq. (6) and further assuming that the perturbation

function $o(x, y)$ satisfies that $|\nabla o| \approx 0$ and $|\nabla^2 o|$ is bounded from above in the domain Ω . Then Eq. (6) is converted into

$$\begin{aligned}\hat{\mu}(o; w) &= \mu(w) + \frac{(1+w_x^2)o_{yy} - 2w_x w_y o_{xy} + (1+w_y^2)o_{xx}}{2(1+w_x^2 + w_y^2)^{3/2}} \\ \hat{\kappa}(o; w) &= \frac{(w_{xx} + o_{xx})(w_{yy} + o_{yy}) - (w_{xy} + o_{xy})^2}{(1+w_x^2 + w_y^2)^2}.\end{aligned}\quad (8)$$

Here, $w(x, y) = \sqrt{R^2 - x^2 - y^2}$ and R is the radius of curvature of the spherical background surface. Replacing μ and κ in Eq. (7) with $\hat{\mu}$ and $\hat{\kappa}$, Eq. (7) changes to the following form:

$$I(o; w) = \int_{\Omega} \{(\alpha + \beta)\hat{\mu}^2 - 2\beta\mu_0\hat{\mu} - \alpha\hat{\kappa} + \beta\mu_0^2\} dx dy. \quad (9)$$

Subsequently, inserting Eq. (8) into Eq. (9) and removing constant terms, a quadratic error functional is formulated by [11]

$$\begin{aligned}I(o; w) &= \int_{\Omega} (a_1 o_{xx}^2 + a_2 o_{xy}^2 + a_3 o_{yy}^2 + a_4 o_{xx} o_{yy} + a_5 o_{xy} o_{yy} \\ &\quad + a_6 o_{xx} o_{xy} + a_7 o_{xx} + a_8 o_{xy} + a_9 o_{yy}) dx dy.\end{aligned}\quad (10)$$

Here, the coefficient a_i is a function with respect to $(w, \alpha, \beta, \mu_0)$, and its specific form can be found in Eq. (2.14) of the work by Jiang et al. [11]. Under the necessary condition for $I(o; w)$ to be minimum, through integration by parts twice, we can get a linear fourth-order PDE with respect to the perturbation term $o(x, y)$:

$$\begin{aligned}(2a_1 o_{xx})_{xx} + (2a_2 o_{xy})_{xy} + (2a_3 o_{yy})_{yy} + (a_4 o_{yy})_{xx} \\ + (a_4 o_{xx})_{yy} + (a_5 o_{yy})_{xy} + (a_5 o_{xy})_{yy} + (a_6 o_{xy})_{xx} \\ + (a_6 o_{xx})_{xy} + (a_7)_{xx} + (a_8)_{xy} + (a_9)_{yy} = 0.\end{aligned}\quad (11)$$

For a desired power distribution $u_0(x, y)$, two positive weight functions $\alpha(x, y)$ and $\beta(x, y)$, and a prescribed radius of curvature R , the above Eq. (11) can be solved with finite difference method to get the perturbation distribution $o(x, y)$. We determine the constant $R = 1000(n - 1)/F_{\text{PAL}}$, where F_{PAL} denotes an optical power value between the far zone and the near zone. The detailed settings and assignments of $u_0(x, y)$, $\alpha(x, y)$ and $\beta(x, y)$ are presented in Section 3.1. The initial estimate of the designed PAL is obtained with $z_0(x, y) = \sqrt{R^2 - x^2 - y^2} + o(x, y)$. Here, we utilize the *sparse.linalg.spsolve* function from the *Scipy* package in Python to solve the PDE, with the premise of transforming the coefficient matrix into a sparse matrix.

2.3. Optimize PAL with NURBS surface

2.3.1. NURBS freeform surface

Differentiable ray tracing methods have been widely applied in the design of illumination and imaging optics, with several approaches utilizing B-spline or its variants to represent freeform surfaces [13–15]. Non-uniform rational B-spline (NURBS) [16], a subset of B-spline, is highly freeform for expressing both global and local features, which aptly matches the requirements of designing PAL. The tensor product of two NURBS curves, parameterized by two parameters u and v , yields a NURBS surface, as formulated by

$$\mathbf{Z}(u, v) = \sum_{i=0}^{m_s-1} \sum_{j=0}^{n_s-1} N_{i,p}(u) N_{j,q}(v) w_{i,j} \mathbf{P}_{i,j}, \quad 0 \leq u, v \leq 1, \quad (12)$$

where $\mathbf{P}_{i,j} = (x_p, y_p, z_p)$ is the control point with weight $w_{i,j}$, m_s and n_s are the number of control points along two non-parallel directions. $N_{i,p}(u)$ and $N_{j,q}(v)$ denote the basis functions of B-spline

of order of p and q that are defined in the u and v directions respectively, which can be given by the De Boor-Cox recursion formula [16]:

$$\begin{cases} N_{i,0}(u) = \begin{cases} 1 & \text{if } u_i \leq u < u_{i+1} \\ 0 & \text{otherwise} \end{cases}, \\ N_{i,p}(u) = \frac{u - u_i}{u_{i+p} - u_i} N_{i,p-1}(u) + \frac{u_{i+p+1} - u}{u_{i+p+1} - u_{i+1}} N_{i+1,p-1}(u), \\ \text{define: } \frac{0}{0} = 0. \end{cases} \quad (13)$$

Here, we apply the quasi-uniform node u_i which divides the interval $[0,1]$ into $m_s + p$ segments. The coordinates along x and y axes of control point $P_{i,j}$ are sampled uniformly. Therefore, the only optimized parameter is coordinate z_p , while x_p and y_p remain fixed. Simultaneously, the expression of NURBS surface can be rewritten by replacing $Z(u, v)$ with $Z(x, y)$. In this paper, we employ the analytical form of the B-spline basis function of degree 2 [13,17], which improves the speed of the interpolation process in optimization. Then we apply gradient descent (GD) algorithm to accomplish the fitting of initial estimate of PAL, by minimizing the merit function

$$M(z_p) = \|Z(x, y) - z_0(x, y)\|_F^2, \quad (14)$$

where $\|\cdot\|_F$ denotes the Frobenius norm of a matrix.

2.3.2. Minimize error functional with NURBS surface

After defining the representation of sag distribution of PAL, all of the derivatives ($z_x, z_y, z_{xx}, z_{xy}, z_{yy}$) shown in Eq. (6) can be easily and efficiently obtained using AD. Compared to numerical differentiation techniques such as finite difference method and finite element method, AD is free from truncation and round-off errors, thereby providing significantly higher accuracy. The diagram of the proposed differentiable optimization method is shown in Fig. 3. For a given power distribution $P_0(x, y)$ and weight distributions $\alpha(x, y)$ and $\beta(x, y)$, the final error functional $I(z)$ in Eq. (7) can be constructed with calculated mean curvature $\mu(x, y)$ and Gauss curvature $\kappa(x, y)$. Furthermore, the GD algorithm is applied for minimizing the error functional.

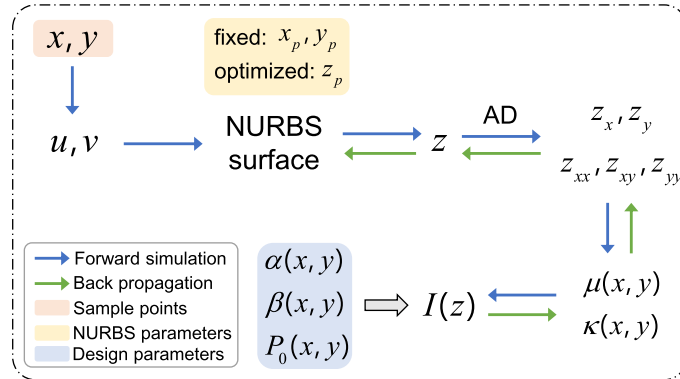


Fig. 3. Diagram of the proposed differentiable optimization method.

Since the entire design process involves iterative refinement with NURBS surface, the weight functions in Eq. (7) can be assigned flexibly to meet the customized requirements (the size of the far zone and near zone, the length and width of the transition zone, the weight ratio between the desired power and the unwanted astigmatism, etc.).

3. Results

To illustrate the effectiveness of the proposed design method, we present two design examples in this section. Due to the inherent limitation of direct design methods in controlling astigmatism, we only compare the quality of the designed PALs against current indirect design methods. In this paper, we take the variational-difference numerical method [11] for contrast. The design domain of the PAL is defined as a square region $\Omega = \{(x, y) \mid -40 \text{ mm} \leq x, y \leq 40 \text{ mm}\}$, while the practical evaluation area is a circular region $\Lambda = \{(x, y) \mid x^2 + y^2 \leq 30^2 \text{ mm}^2\}$. The refractive index of the PALs is 1.530. All the design and simulation procedures are conducted on a CPU of Intel of i9-10900K 3.70 GHz, a GPU of NVIDIA RTX 3080, a windows 10 system. The AD is conducted by PyTorch in version 1.10.2. The optimizer for GD algorithm is Adam [18].

3.1. Design parameters assignment

The first step of the design method is to assign the effective visual region (far zone, near zone and intermediate zone). Following the procedure used by Wang et al. [10] and Jiang et al. [11], the design domain is divided into seven regions, as indicated by the solid black lines in Fig. 4(a). Then we assign the weight distributions $\alpha(x, y)$ and $\beta(x, y)$ according to the partition of the PAL. The initial value of weights in effective visual region is set to 30.0; the initial value of weights in blending zone is set to 1.0. Furthermore, within the circle region Λ , the weight value follows the previous definition, while outside the circle, it is set to a minimum value (e.g. $\approx 10^{-8}$). Finally, a convolution operation is applied to smooth the segmented weight functions, which affects the smoothness of the solution. Here, the preliminary distributions of the two weight functions are identical, as illustrated in Fig. 4(b). In our program tests, when the weight ratio between the effective visual region and the blending zone exceeds approximately 1000:1, the maximum astigmatism on both sides of the progressive corridor exceeds one times the addition power.

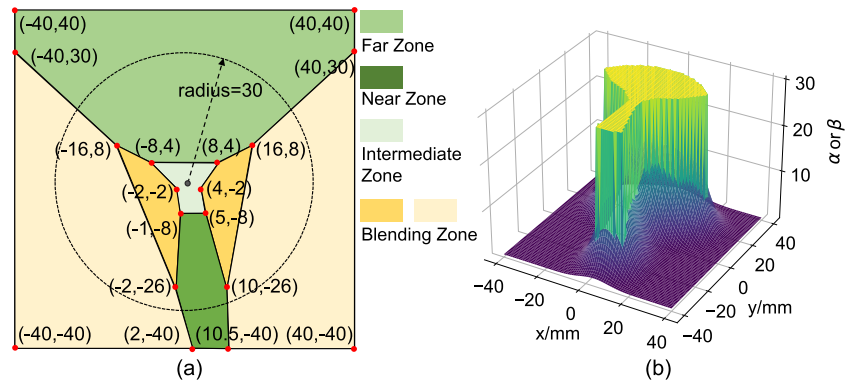


Fig. 4. (a) Partition of the square design domain for PAL (unit : mm). (b) Distribution of the weight function $\alpha(x, y)$ or $\beta(x, y)$.

The desired power distribution $P_0(x, y)$ depends on the wearer's individual prescription, which is determined by the base power and addition power. The base power provides the distance vision while the addition power is the power difference between the far zone and the near zone. For a patient suffering from both presbyopia and myopia, the combination of the front and back surfaces contributes to the final correction. Assuming that the front surface is a PAL and the back surface is spherical, the power for correcting presbyopia is equal to the addition power of the PAL. The myopia correction is achieved by combining the base power of the PAL with the spherical power of the back surface. We have designed two different prescriptions identical to those used in the work by Jiang et al. [11], as shown in Table 1. We assign φ_{Base} and $(\varphi_{\text{Base}} + \varphi_{\text{ADD}})$ to the far and

near zones respectively, while the progressive transition in the intermediate zone is achieved using a sine (or cosine) function. The remaining power of the PAL is set to $(\varphi_{\text{Base}} + (\varphi_{\text{Base}} + \varphi_{\text{ADD}}))/2$. After a convolution operation, the smooth prescribed power distributions are shown in Fig. 5. We then replace the optical power with mean curvature $\mu(x, y)$ using Eq. (3).

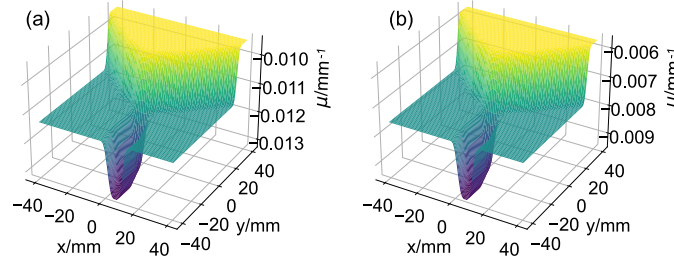


Fig. 5. Prescribed mean curvature distributions. (a) Example 1; (b) Example 2.

Table 1. Two design prescriptions of PAL^a

Example	Myopia	Presbyopia	φ_{Base}	φ_{ADD}	φ_{Sphere}
1	-2.0	2.0	5.0	2.0	7.0
2	-1.5	2.0	3.0	2.0	4.5

^aThe φ_{Base} and φ_{ADD} are the base power and addition power of PAL, the φ_{Sphere} is the power of the back surface, all of the design parameters are measured in diopter (D).

3.2. Optimizing results

The detailed optimization processes of the two examples are as follows. Before optimizing the NURBS surface, an initial estimate of PAL is generated by solving a linear fourth-order PDE with respect to the perturbation term $o(x, y)$, as shown in Eq. (11). The radii of the spherical background surface $w(x, y)$ are set to 106 mm and 140 mm for Example 1 and Example 2, respectively. The number of computational grids is 81×81 , with an interval of 1 mm. Coupled with the boundary constraints $o(-40, -40) = o(40, -40) = o(40, 40) = 0$, the numerical results obtained using the finite difference method are presented in Fig. 6. We chose 30×30 control points to fit the obtained discrete points, as further increasing the number leads to only marginal improvements in fitting accuracy.

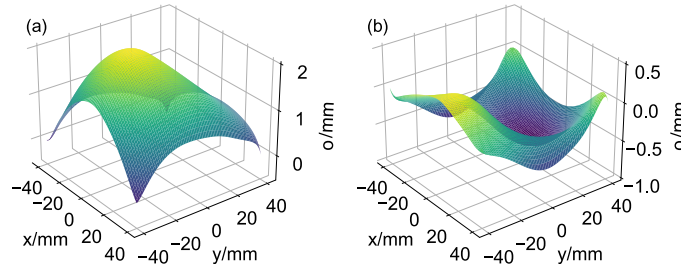


Fig. 6. Numerical results for perturbation term $o(x, y)$. (a) Example 1; (b) Example 2.

In the optimization for error functional using NURBS surface, We initially explored the impact of varying weight ratios between the desired power and the unwanted astigmatism. We optimized

30 groups with different weight distributions for optical power and astigmatism respectively, as shown in Fig. 7. The value of addition power (also represented as ADD) is calculated from the reference distance point (0, 5) to the reference reading point (2, -9). The length of the corridor (or the intermediate zone shown in Fig. 4(a)) is 12 mm and the width of the corridor is between 6 mm and 16 mm. The value of astigmatism represents the average astigmatism across the effective visual area and blending zone within the circular region Λ . We can observe that there is a linear relationship between the addition power and the astigmatism, as indicated by the green line. When the weight ratio continues to increase (the ratio of the maximum value of $\beta(x, y)$ to that of $\alpha(x, y)$ is approximately 5:3), the high-level astigmatism areas gradually shift and concentrate along the lateral areas of the progressive corridor. Ultimately, the progressive corridor is closed with $P_{ast} > 0.25$ D (red arrow in Fig. 7). Our final design weight distribution is shown in blue arrow in Fig. 7. The initial learning rate of GD algorithm is 5×10^{-3} , with a decay rate of 0.975 every 50 iterations.

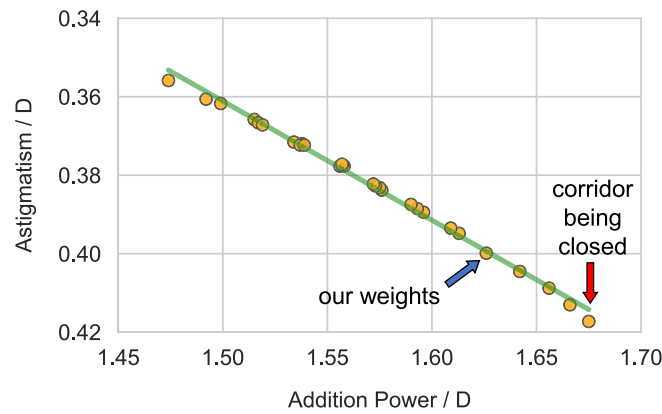


Fig. 7. Variation trend between the astigmatism and the addition power across different weight ratios, with 30 optimized groups.

The evolution curve of the error functional $I(z)$ with respect to iterations is shown in Fig. 8. Following 1300 iterations optimizing the NURBS surface (a total of 1.55 minutes, or about 0.0715 seconds per iteration), the error functional gradually approaches a stationary value. Through additional convolution operation, the second-order NURBS surface is at least third-order continuous and differentiable. The sag and curvature distributions of the optimized NURBS surfaces are shown in Fig. 9. To evaluate the optical properties of the final designed PALs, contour plots of sphere power and cylinder power (astigmatism) for the two examples are presented in Fig. 10. We have uniformly sampled 81×81 points on the NURBS surface, as expressed by Eq. (12) and Eq. (13). The contour lines are at intervals of 0.25 D, and the lens diameter used for plotting is 60 mm. Specifically, sphere power and cylinder power are determined by Eq. (3), while mean curvature and curvature difference are readily obtained from Eq. (6). To analyze the design results of Example 1, the cylinder and sphere power of linear solution are shown in Fig. 10(a) and Fig. 10(b). It can be observed that the astigmatism value remains below 0.50 D across most of the effective visual region, with an addition power of 1.499 D. After the differentiable optimization using NURBS surface, astigmatism decreases further below 0.25 D in the effective visual region, and the addition power increases to 1.625 D. Compared to linear solution, the astigmatism value in the effective visual region is below 0.25 D and the addition power increases by 8.41 %. Although the maximum value of astigmatism in the blending zone is increasing, the high-level astigmatism areas near the intermediate transition zone are pushed towards the edge of the lens, an area that is ineffective for wearers. For the design results of Example 2, the

astigmatism in the effective visual region decreases below 0.25 D from a linear solution mostly within 0.50 D. The addition power increases from 1.456 D to 1.626 D, a rise of 11.68%. Similarly, the high-level astigmatism areas shift to both sides of the lens, confined to a small region. During the optimization, We also find that 30×30 control points are sufficient for the optimization of the error functional, and increasing the number further results in a mutually constrained effect between minimizing the astigmatism and making the addition power closer to the target value. To illustrate with the optimization result of Example 1, the maximum astigmatism of PAL optimized with 30×30 control points is 1.834 D and the addition power is 1.625 D. When optimizing with 35×35 control points, the maximum astigmatism is 1.789 D (a rise of 0.045 D) and the addition power is 1.615 D (a decrease of 0.01 D).

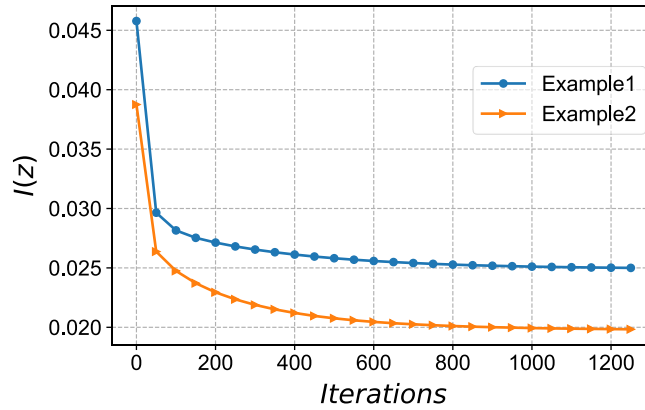


Fig. 8. Evolution curve of the error functional $I(z)$ during the optimization process.

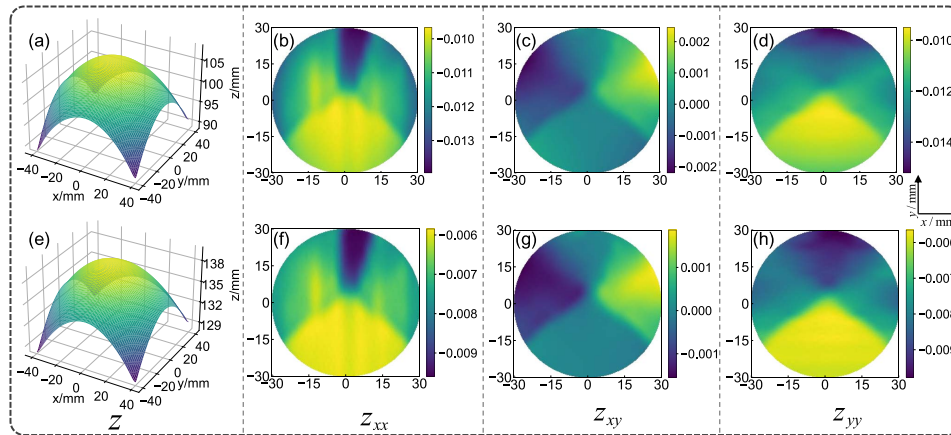


Fig. 9. Sag and curvature distributions of optimized NURBS surfaces. Example 1: (a) sag z ; (b) z_{xx} ; (c) z_{xy} ; (d) z_{yy} . Example 2: (e) sag z ; (f) z_{xx} ; (g) z_{xy} ; (h) z_{yy} .

Considering the application of generating a PAL, we have explored the impact of decentration and tilt on its final performance. The tilt of the surface along the x -axis or y -axis is set to $\pm 5^\circ$, corresponding to the angle between the visual axis and the optical axis of the human eye. The decentration of the surface in the x or y directions is set to ± 2 mm. The above disturbances generated based on uniformly distributed random numbers are randomly and independently applied to the lens. We define three metrics to evaluate the performance of the PALs, including

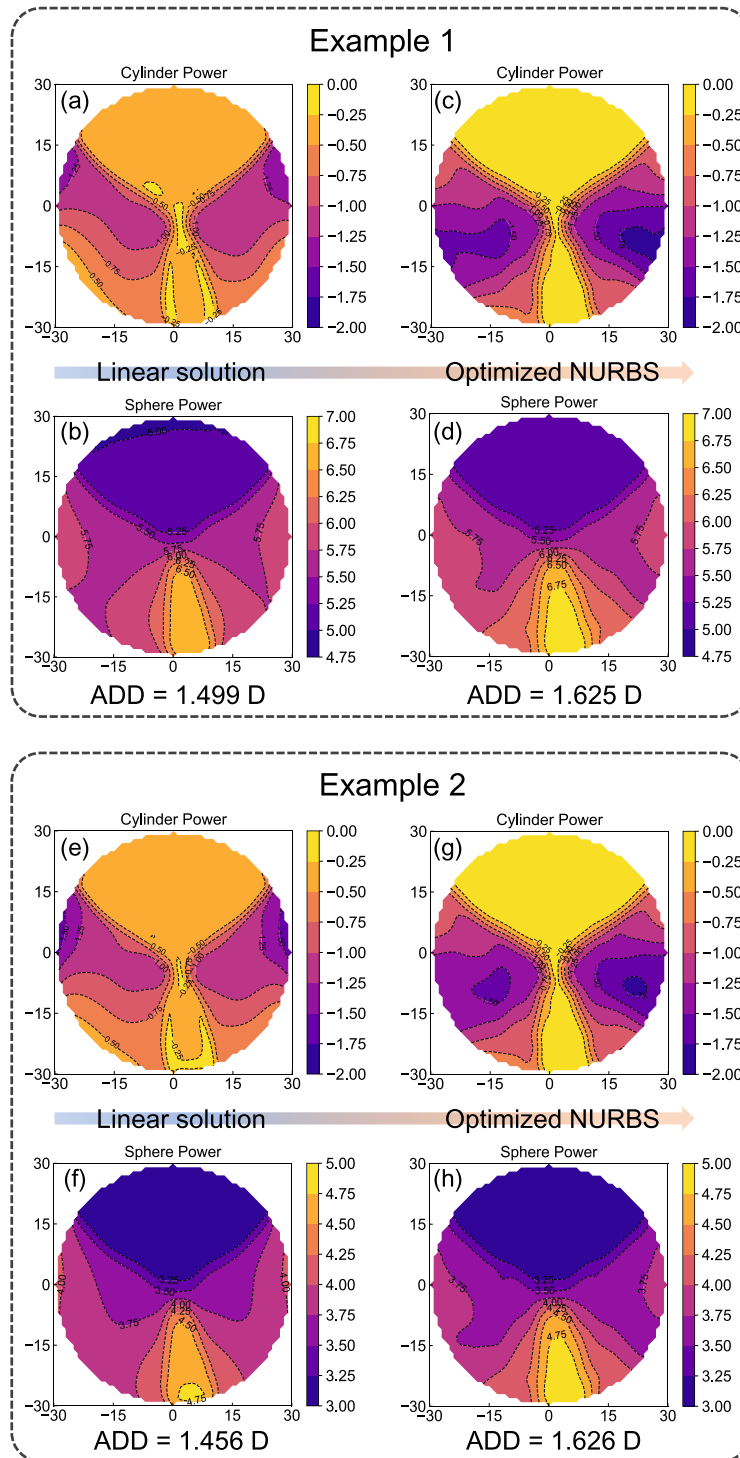


Fig. 10. Contour plots of the optimized results for Example 1 and Example 2. Cylinder power : (a) (e) linear solution [11]; (c) (g) optimized solution using NURBS surface. Sphere power : (b) (f) linear solution [11]; (d) (h) optimized solution using NURBS surface.

the mean value of astigmatism in the effective visual region, the maximum value of astigmatism on both sides of the progressive corridor and the mean value of optical power error in the effective visual region, denoted by $\text{Mean-}P_{\text{ast}}$, $\text{Max-}P_{\text{ast}}$ and $\text{Mean-}P_{\text{m-Error}}$, respectively. The results of Monte Carlo simulation for Example 1 and Example 2 are shown in Table 2 and Table 3, respectively. We can find that at least 90% of the simulation results meet the design requirements for both examples. The mean astigmatism in the effective visual region is less than 0.25 D, and the maximum astigmatism on both sides of the progressive corridor is less than the addition power 2 D. Additionally, the mean optical power error in the effective visual region is less than 0.15 D.

Table 2. The result of Monte Carlo simulation for the decentration and tilt of Example 1, a total of 1000 simulations.

	Mean- P_{ast} (D)	Max- P_{ast} (D)	Mean- $P_{\text{m-Error}}$ (D)
98%	<0.276	<1.801	<0.164
90%	<0.249	<1.763	<0.143
80%	<0.230	<1.730	<0.127
50%	<0.180	<1.649	<0.094
20%	<0.146	<1.584	<0.068
10%	<0.131	<1.557	<0.058
2%	<0.114	<1.510	<0.047

Table 3. The result of Monte Carlo simulation for the decentration and tilt of Example 2, a total of 1000 simulations.

	Mean- P_{ast} (D)	Max- P_{ast} (D)	Mean- $P_{\text{m-Error}}$ (D)
98%	<0.225	<1.735	<0.117
90%	<0.199	<1.702	<0.104
80%	<0.186	<1.682	<0.095
50%	<0.157	<1.637	<0.075
20%	<0.130	<1.601	<0.059
10%	<0.121	<1.583	<0.052
2%	<0.113	<1.547	<0.043

4. Conclusion

We have proposed a differentiable optimization method using NURBS surface for PAL design. In this method, we minimize a nonlinear error functional instead of solving a linear fourth-order PDE directly. To obtain an initial estimate of PAL, we use the finite difference method to solve a linear fourth-order PDE involving the perturbation term based on a spherical background surface. Then we employ NURBS surface to represent the sag distribution of PAL, minimizing the error functional with GD algorithm. Compared to existing design methods, the aforementioned two design examples demonstrate the effectiveness of the proposed approach in reducing astigmatism within the effective visual area and achieving an addition power closer to the target value. More importantly, the high-level astigmatism areas near the intermediate transition zone are pushed towards the lens edge. Our method offers a more flexible adjustment around the linear solution to achieve satisfactory design result, as the linear solution is practically justified, given that the surface of a progressive lens is nearly spherical for most prescriptions.

The proposed design method can also accommodate the personalized demands, with the desired power distribution, domain partitioning, and weight function distributions all tailored to

the wearer's requirements. The trade off between the desired power and the unwanted astigmatism is adjustable flexibly, based on the visualization outcomes from different optimization iterations. Our design method may provide a more acceptable and comfortable correction scheme for the presbyopic wearers. In future work, we will consider using differentiable ray tracing technique to design the PAL with double freeform surfaces and further enhance its correction capability.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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